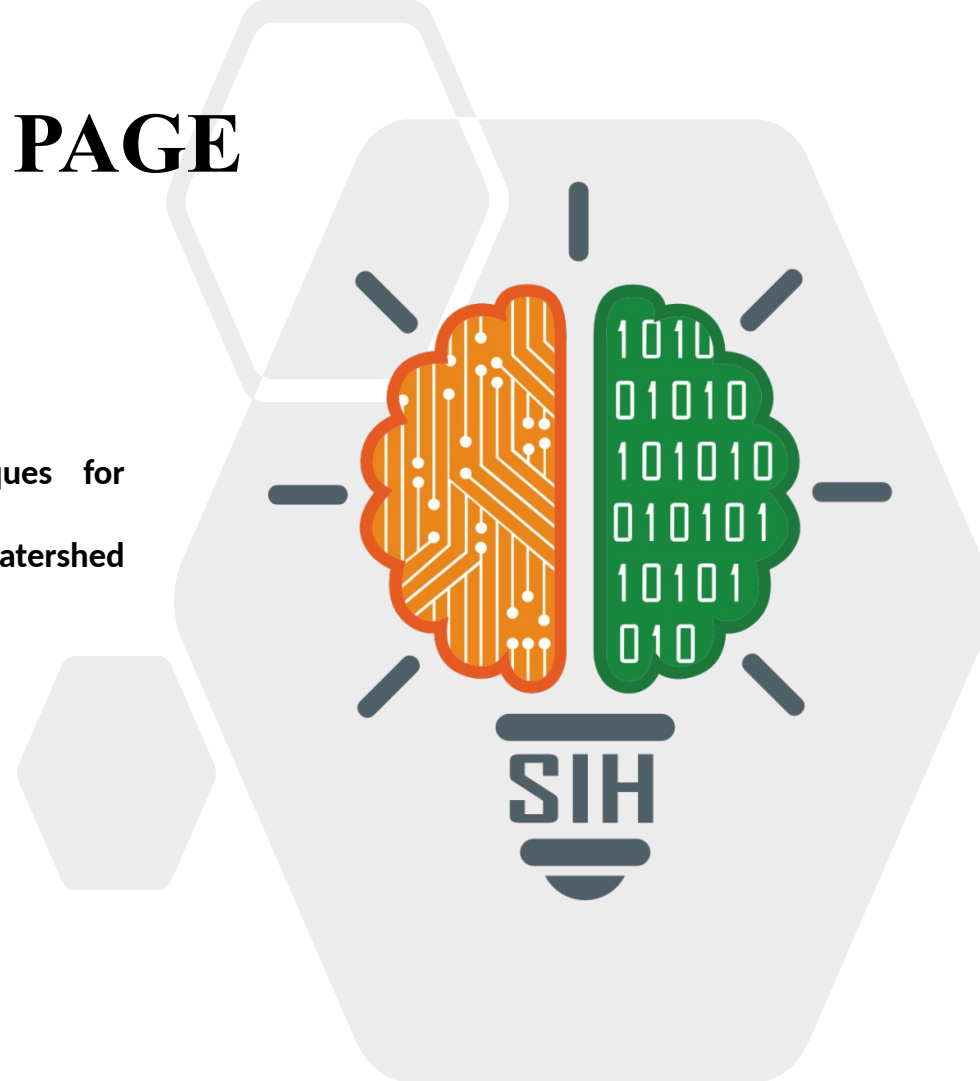


SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID - 26015
- Problem Statement Title - Application of Geospatial Techniques for visualization and analysis to interpret Geo-Coded Images to enhance watershed Development Outcomes
- Theme - Agriculture, FoodTech & Rural Development
- PS Category - Software
- Team ID - <your team ID>
- Team Name (Registered on portal) - <your team name>



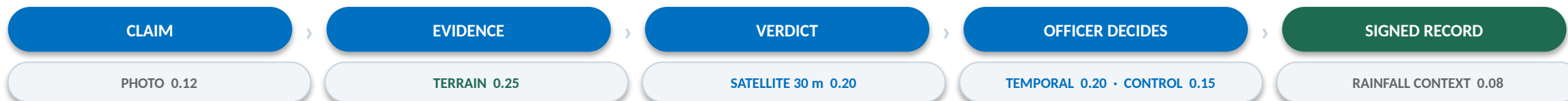
PRAMAAN • प्रमाण

Turning geo-tagged watershed photographs into verifiable evidence.

IDEA TITLE

Proposed Solution (Describe your Idea/Solution/Prototype)

PRAMAAN — an evidence-reconciliation layer over DoLR's existing SRISHTI / DRISHTI geotag pipeline. Every field photograph becomes a claim the satellite record can test.



Detailed explanation of the proposed solution

- Each DRISHTI geotag — lat/lon, GPS accuracy, orientation, timestamp, activity, status — becomes a structured, testable claim
- Reconciled against six evidence families — terrain, satellite, temporal, matched controls, rainfall context and the photograph. Five are independent of the claim; the photograph is not, which is why it is weighted lowest
- Output: a verdict on a published L0–L4 evidence ladder, with calibrated confidence and a mandatory dissent panel
- Officer accepts / edits / rejects; signed into an append-only, hash-chained ledger
- Deliverable: an Evidence Pack PDF carrying every scene ID, model version and weight used

How it addresses the problem

- Today a geotag is moderated by eye — accept or reject. That verifies a photograph, not an outcome
- 1,24,830 water-harvesting structures under WDC-PMKSY 2.0 cannot be eyeballed at scale
- Implements the WDC-PMKSY 2.0 written mandate: NDVI/NDWI change detection, and cross-verification of satellite images with ground interventions
- Converts ~1,200 works per district into ~40 that need a human decision, each with a documented reason
- Adds one step to the existing workflow — replaces no government system

Innovation and uniqueness of the solution

- Matched-control differencing per structure — 5–12 similar untreated sites in the same sub-watershed, so we measure the intervention, not the monsoon
- Terrain plausibility screen — a check dam claimed off any drainage line is flagged deterministically, before any AI runs
- Detectability gate — the system refuses to judge structures smaller than one 30 m pixel and escalates to a cluster claim
- Causal claims are unreachable in code; “requires physical verification” is the strongest language it can emit
- Every officer correction becomes labelled training data

TECHNICAL APPROACH



Technologies to be used (e.g. programming languages, frameworks, hardware)

BACKEND

Python 3.11 • FastAPI • Celery • Redis • Docker Compose

GEOSPATIAL

PostgreSQL + PostGIS • GDAL • rasterio • GeoPandas • Shapely • PyProj • WhiteboxTools

AI / CV

PyTorch • SigLIP-2 zero-shot (no labelled corpus exists) • OpenCV • Prithvi-EO-2.0 30 m foundation model as the research path

FRONTEND

React • TypeScript • MapLibre GL • Tiler for COG tiles

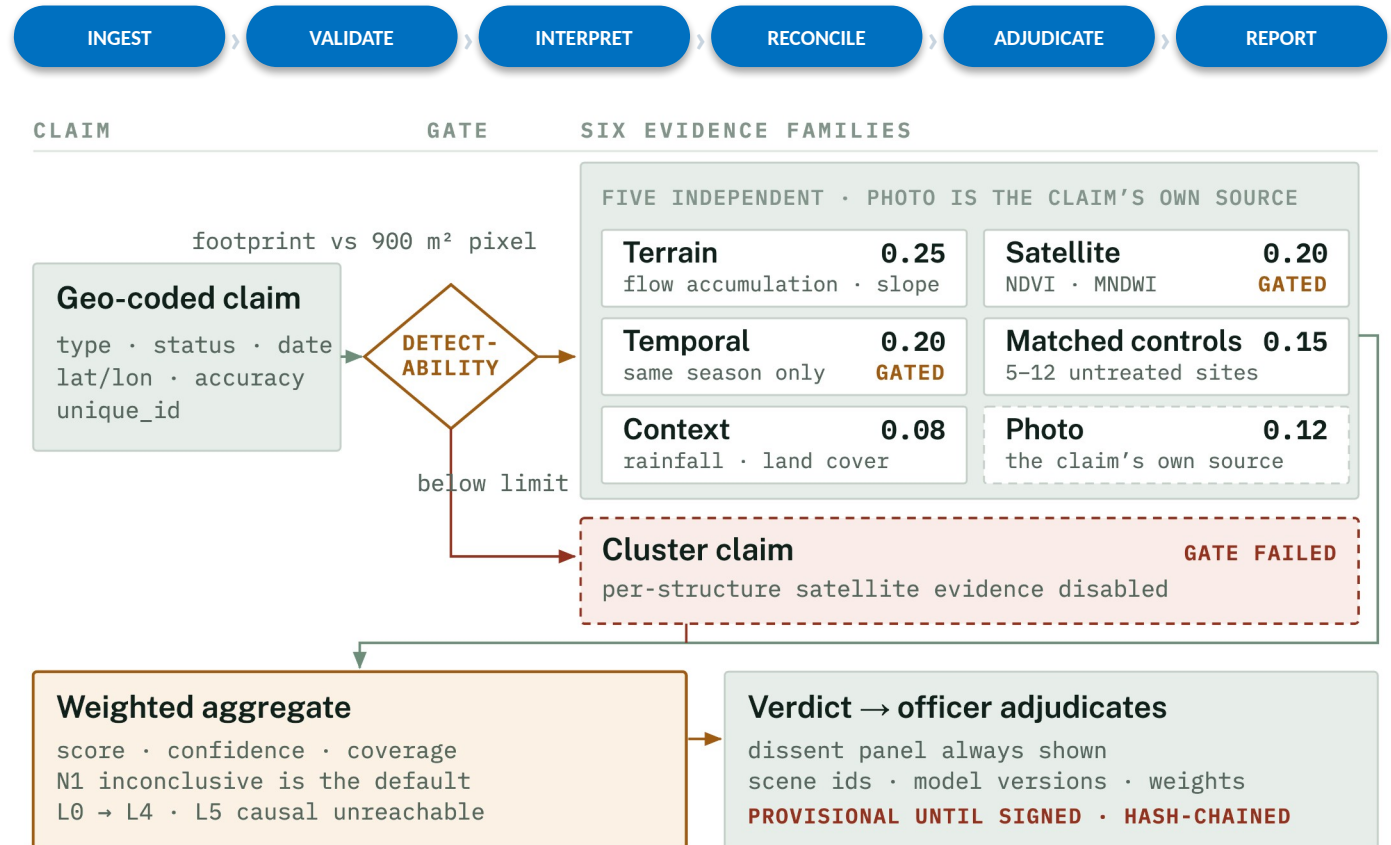
DATA SOURCES

NASA HLS 30 m (STAC) • Copernicus Sentinel-2 (STAC) • NRSC Bhoonidhi API (JWT + STAC) • Bhuvan OGC WMS • JRC Global Surface Water • CartoDEM • SLUSI watershed atlas

HARDWARE

One 8-core VM. No GPU required on the core path. ₹0 in data and licence cost.

Methodology and process for implementation (Flow Charts / Images / working prototype)



Terrain, satellite indices and temporal statistics are deterministic — no machine learning in three of the six families.

The reconciliation mechanism. Six evidence families weighted into one verdict — five independent of the photograph, which is the claim's own source and weighted lowest. The detectability gate disables per-structure satellite evidence below one 30 m pixel; the engine is a pure function — same bundle, same verdict.

FEASIBILITY AND VIABILITY



Analysis of the feasibility of the idea

- Every core data source is free, open and programmatic — endpoints verified, not assumed
- NRSC Bhoonidhi exposes a documented JWT + STAC API; data coarser than 5 m is open under Indian Space Policy 2023
- ₹0 in data and licence cost; entire stack is open-source
- Runs on a single 8-core VM with no GPU on the core path
- Built to NRSC's published DRISHTI field schema, so production integration is a driver swap, not a rewrite
- Complete workflow demonstrable offline from a local cache

Potential challenges and risks

- Many structures are smaller than one 30 m pixel — farm ponds, gully plugs, individual trenches
- Monsoon cloud cover can destroy an entire kharif observation window
- EXIF GPS is often stripped by messaging apps; coordinates may be typed manually
- Seasonality and rainfall can be mistaken for genuine intervention impact
- A false negative could unfairly implicate a beneficiary
- No public SRISHTI / DRISHTI API could be found during research

Strategies for overcoming these challenges

- Detectability gate compares footprint to sensor resolution first, then escalates to a cluster claim — never a false negative
- Rabi and summer windows carry the analysis; cloud is scored per area of interest, not per scene
- Ranked provenance chain (EXIF → sidecar → CSV → manual pin) plus a deterministic terrain plausibility screen
- Matched controls in the same sub-watershed, same-season comparison, construction window excluded
- Verdicts stay provisional until a named officer signs; wording is locked to "requires physical verification"
- Imagery and geotag sources sit behind driver interfaces — one class to swap

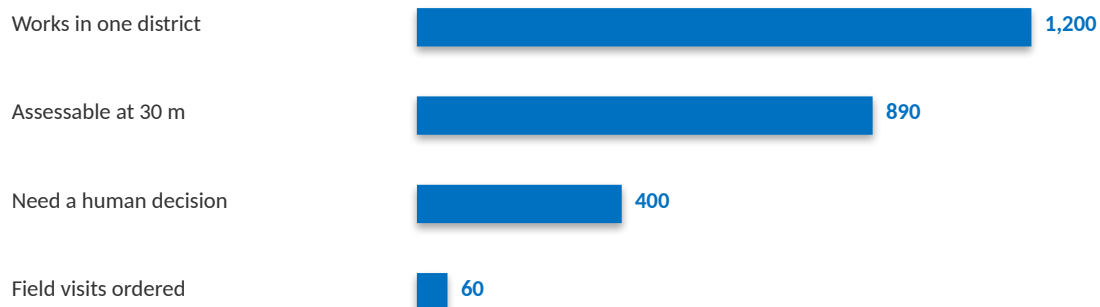
IMPACT AND BENEFITS



Potential impact on the target audience

- WDCDC Project Manager — ~1,200 works per district reduce to ~40 needing a decision each month, each with a documented reason for the field visit
- SLNA monitoring & evaluation — a standing, reproducible evidence pipeline replaces hand-built one-off GIS analyses
- DoLR — every aggregate number traces to a signed adjudication, making progress reporting audit-defensible
- Field WDT — fewer wasted verification journeys; feedback on capture quality

TARGETING SCARCE VERIFICATION EFFORT — ONE DISTRICT, ONE MONTH



Benefits of the solution (social, economic, environmental, etc.)

ECONOMIC

- Targets a scarce verification budget inside a ₹8,134 crore programme covering 49.5 lakh hectares
- ₹0 data and licence cost — marginal cost per extra structure is CPU seconds

SOCIAL

- Accountability without accusation — the system never says fraud or false
- Strongest wording it can emit is “requires physical verification”; a named officer makes the finding

ENVIRONMENTAL

- Earlier detection of non-performing soil and water conservation works
- Faster remedial action on degraded land, water retention and vegetation cover

INSTITUTIONAL

- An expert-labelled dataset of Indian watershed field evidence grows from routine use
- Owned by the department, at no additional cost

Watershed development is already proven: mean benefit-cost ≈ 2 , IRR 27.4%, cropping intensity +35.5% (ICRISAT, 636 Indian micro-watersheds). The gap is knowing which structures work.

RESEARCH AND REFERENCES



Details / Links of the reference and research work

GOVERNMENT OF INDIA — PROGRAMME & PLATFORM

- WDC-PMKSY 2.0 portal and national dashboard — wdcpmkxy.dolr.gov.in
- Guidelines for New Generation Watershed Development Projects (WDC-PMKSY 2.0), DoLR — mandates NDVI/NDWI change detection and cross-verification of satellite images with ground interventions
- National Technical Guidelines for Improved Watershed Management, NRAA, Aug 2025
- NRSC SRISHTI-DRISHTI User Manual and DRISHTI v2.3 Manual — bhuvan-app1.nrsc.gov.in/iwmp
- Bhuvan Watershed Development Component 2.0 — bhuvan-app1.nrsc.gov.in/wdc2.0
- MoRD press release, DoLR-NRSC MoU, 18 Oct 2023 — SRISHTI 2.0 and DRISHTI 2.0
- PIB: one crore MGNREGA assets geotagged (PRID 1488368)

DATA SOURCES & SCIENTIFIC LITERATURE

- Bhoonidhi API Specification, NRSC — bhoonidhi.nrsc.gov.in/bhoonidhi-api (JWT + STAC)
- NASA Harmonized Landsat Sentinel-2 (HLS) 30 m v2.0 — LP DAAC
- Copernicus Data Space Ecosystem STAC catalogue
- JRC Global Surface Water v1.4 (Pekel et al.) — 30 m water history, 1984–
- SLUSI Digital Watershed Atlas of India
- Prithvi-EO-2.0: A Versatile Multi-Temporal Foundation Model for Earth Observation — arXiv:2412.02732 (pretrained on 4.2 M HLS samples at 30 m)
- Assessing impacts of watershed interventions using ground data and remote sensing, Ethiopia — Springer, doi 10.1007/s13762-021-03192-7
- Impact of Watershed Program and Conditions for Success: A Meta-Analysis — ICRISAT, 636 micro-watersheds
- Small water bodies mapped from Sentinel-2 MSI — Int. J. Remote Sensing 41(20)